

ed framework. That means implementation is quicker. That's a big plus compared to the more complex reactor types that are still under development. "There has been an overwhelming response to the RFP, and we look forward to a great engagement with our industry partners to meet the shared goal of decarbonisation," says Pathak.

The bidding process is nearing completion, and the government will announce the successful bidders on May 2. NPCIL will build, own, and operate the plants, with companies putting in the capital investment. The companies will have the right to use the nuclear power generated for their captive use. They may sell excess power at a pre-agreed tariff. Nearly 50% of the 100 GW nuclear capacity is expected to come from private players, according to the Department of Atomic Energy.

This is a huge opportunity for private players, both in terms of capacity addition and investments. Also, nuclear power tariff is competitive with renewables at around ₹5 per MW, especially when considering the added costs of storing renewable energy for use when the sun isn't shining or the wind isn't blowing.

In November 2021, during the COP26 summit in Glasgow, Prime Minister Narendra Modi had announced that India was committed to achieving net-zero carbon emissions by 2070 and that nuclear energy would play a key part in that transition. This fits with India's broader energy and climate goals—expanding non-fossil fuel energy sources and reducing the dependence on polluting fuels such as coal.

NEED FOR STABLE ENERGY SOURCES

Renewables such as solar and wind have limited capacity due to dependence on weather. This makes it difficult to ensure a consistent supply of energy. Storage technology such as batteries are required to offset this, but the available storage is costly, has low capacity, and degrades fast. The integration of renewable energy sources into the power grid is not without challenges either as the variability of power generation requires utilities to build advanced grids to ensure stability. In addition, renewable energy infrastructure is highly capital- and location-specific in terms of efficiency. These constraints point towards the importance of having secure, consistent sources of energy such as nuclear.

The companies have already started preparatory work on captive nuclear capacity. Tata Power has set up a dedicated team on nuclear power that is looking at global technology options, regulatory approvals, water/fuel requirement and sourcing, technology for fuel re-processing and storage.



PRAVEER SINHA

CMD, Tata Power

“We have looked at numerous potential locations in the country that are good for nuclear power plants in terms of support from local governments and people”



NIKHIL DHINGRA

CEO, ACME Solar

“The cost competitiveness needs to be there, and the commercial construct needs to be solid in terms of the economics and the safety of the investment”

“We have looked at numerous locations that are good for nuclear power plants in terms of support from local governments and people because there should not be any resistance to this type of project,” Tata Power CMD Praveer Sinha tells *BT*.

Naveen Jindal group has announced a new company—Jindal Nuclear Power Private Ltd—with ₹1.80 lakh crore investment over the next two decades and a planned nuclear capacity of 18 GW. Anil Agarwal-led Vedanta Group has floated an expression of interest seeking bids from global firms to build 5 GW nuclear